

Detection of AI Generated Background noise using Deep Learning

CSE 676: Deep Learning

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Problem Statement

We are developing a deep learning system to distinguish between authentic and AI-generated background audio. This system focuses on detecting the subtle artifacts and patterns found in deepfake background noise



Dataset

- EnvSDD** - ~360 hours of environmental audio with synthetic artifacts
- LENS-DF** - long-form audio with complex background noise
- ASVspoof**- Large real-world audio

Methodology

- Balanced Partitioning:** We apply a stratified split with an equal distribution of real and fake samples to ensure unbiased training.
- Robust Augmentation:** Training data is diversified via **time** and pitch shifting to improve model generalization across different acoustic environments.
- Standardized Evaluation:** All architectures are trained within a unified framework

Models

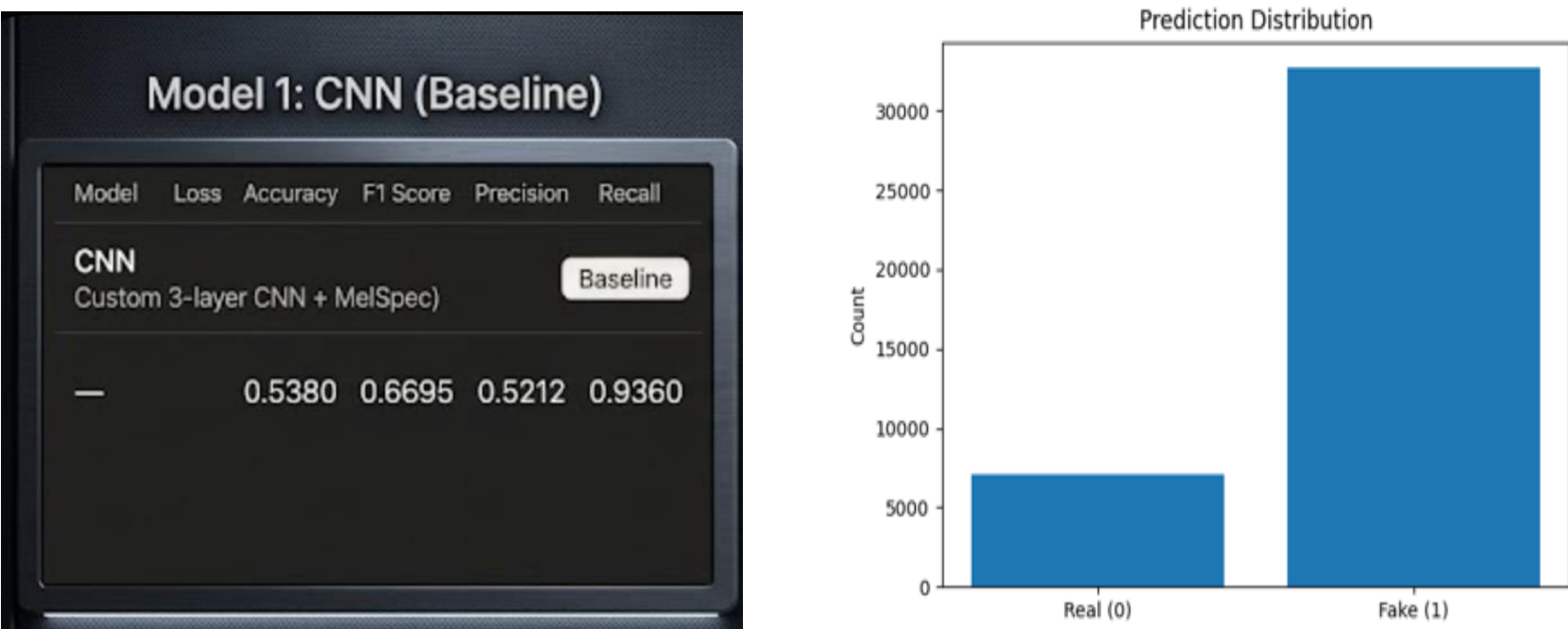


Analysis

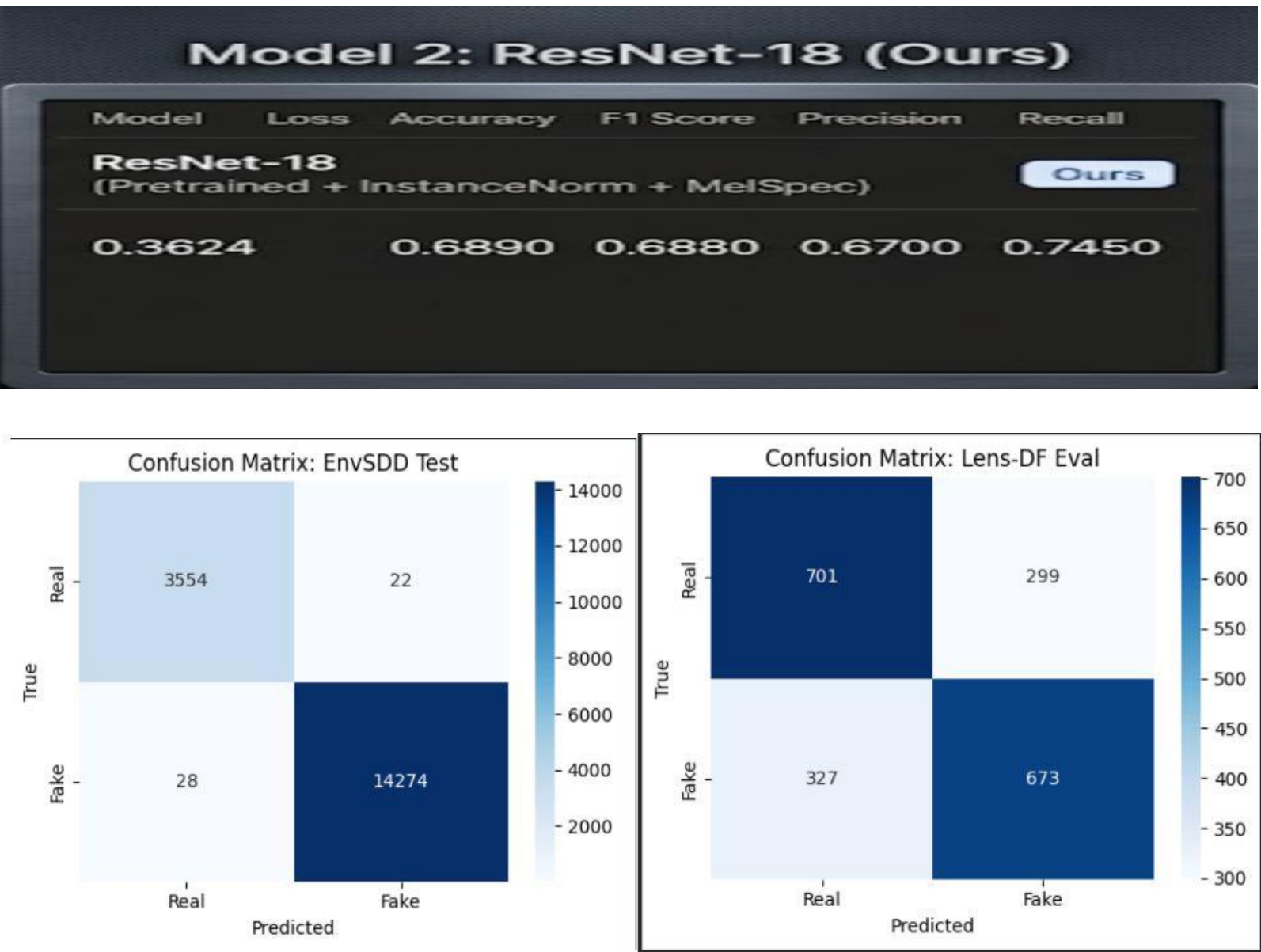
- Fixing Generalization Gaps:** Initial models failed on studio-grade audio after training on environmental noise. We merged datasets to force the system to recognize synthetic patterns across different recording qualities
- Deeper Feature Learning:** Switching to **ResNet** on the combined data allowed the model to use its depth to catch complex artifacts. This reduced false negatives and improved detection of subtle acoustic nuances
- Advanced Waveform Analysis:** We integrated **Wav2Vec2** to handle uncertain predictions in high-fidelity audio. Its specialized embeddings successfully spotted deepfake signatures that traditional spectral methods missed.

Results

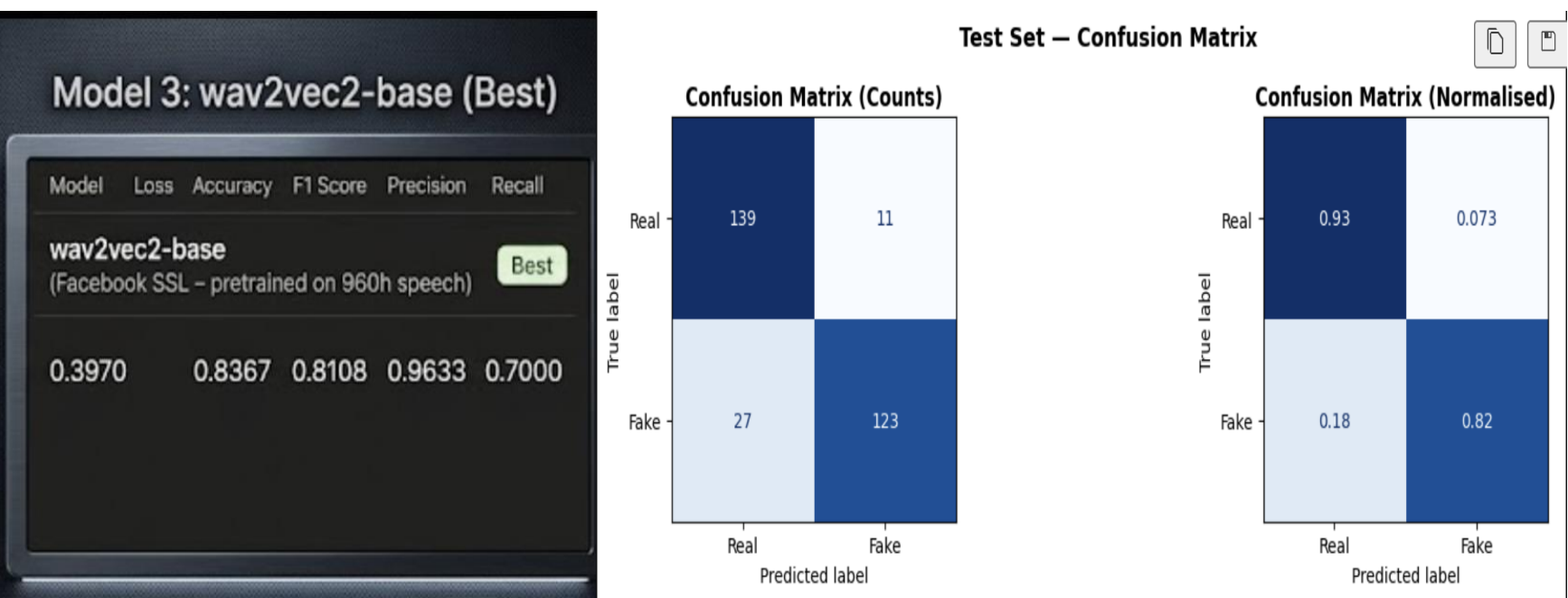
CNN Baseline: Our initial model struggled with generalization, achieving high recall but near-random accuracy. Its shallow architecture couldn't differentiate between real environmental sounds and synthetic artifacts across different recording qualities



ResNet & Data Fusion: We moved to ResNet to leverage deeper layers and merged datasets to bridge the domain gap. This significantly boosted performance on environmental audio but still faced limitations with high-fidelity studio recordings.



Wav2Vec 2.0: We finally transitioned to Wav2Vec2 to extract more sophisticated audio embeddings. This self-supervised approach captured the subtle synthetic signatures in clean studio speech that traditional spectral methods failed to detect.



Conclusion

- Transitioning from basic CNNs to ResNet and Wav2Vec2 proved that merging data with specialized waveform analysis is key to spotting deepfake artifacts. This multi-model approach successfully bridged the gap between noisy environmental sounds and clean studio audio, providing a more robust system for audio authentication

References

- CNN (Baseline):** LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324. <https://doi.org/10.1109/5.726791>
- ResNet:** He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Wav2Vec 2.0:** Baevski, A., Zhou, H., Mohamed, A., & Auli, M. (2020). Wav2Vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems (NeurIPS)*.